

ADEPOJU YUSUF ABIODUN

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CAREER SUMMARY

AI Engineer focused on building and deploying practical LLM applications. I have hands-on experience working with open-source models like Llama and Mistral, specifically in developing RAG pipelines and optimizing prompt performance for production. Beyond just model development, I'm comfortable handling the full lifecycle from data preprocessing in SQL/Python to deploying scalable APIs with Flask. My goal is always to bridge the gap between complex AI research and functional, business-ready tools.

TECHNICAL SKILLS

Generative AI: RAG Pipelines, Prompt Engineering, Embeddings, Vector DBs (ChromaDB/FAISS), Hugging Face, Ollama, Groq.

LLMs: Llama, Gemini, Claude, Mistral, Qwen, GPT-series.

Machine Learning/Deep Learning: Scikit-learn, XGBoost, Model Evaluation.

Programming & Tools: Python (NumPy, Pandas), SQL (PostgreSQL, MySQL), n8n (AI Orchestration), Flask, Git, Power BI.

TECHNICAL PROJECTS

SmartRAG App: Built a RAG system to pull insights from documents so users don't have to search manually.

- Used vector embeddings to ground the LLM's responses in the actual data, which helps stop the model from making things up (hallucinating).
- **Link:** https://github.com/yusufadepoju1/Rag_app

Groq/Llama AI Chatbot

- Developed and deployed a conversational chatbot powered by Llama-3, using Groq to achieve near-instant inference speeds.
- Focused on prompt engineering to maintain dialogue flow and ensure the model handles diverse user queries accurately.
- **Live Demo:** <https://chatbot-3-w1ns.onrender.com/>
- **Code:** <https://github.com/yusufadepoju1/Chatbot>

PDF Data Extraction Tool

- Built a Python utility to automate the extraction of data from unstructured PDFs into clean CSV files.
- Optimized the processing logic to handle high-volume documents, successfully looping through 1,000+ pages in a single session

without memory crashes.

- **Live Demo:** <https://pdfreader4-1.onrender.com>
- **Code:** https://github.com/yusufadepoju1/pdf_reader/

Deep Learning NLP Chatbot:

- Trained a custom NLP model to handle conversational dialogue using deep learning frameworks.
- The project demonstrates the full end-to-end workflow, from text preprocessing and tokenization to final model evaluation.
- **Link:** https://colab.research.google.com/drive/1L8p0q--CF_nTwSz9S3YpAGPis7Eida52

Deep Learning Fake News Classifier

- Developed a supervised deep learning model to detect and classify misinformation within large-scale text datasets.
- Implemented advanced NLP preprocessing techniques, including TF-IDF/Word Embeddings and sequence padding, to improve the model's ability to distinguish between credible and deceptive text.
- Evaluated model reliability using precision-recall curves, focusing on minimizing "false positives" to ensure high-integrity content filtering.
- **Link:** https://colab.research.google.com/drive/13dhU_gVi_AZ1ohKhj7ZzvOuniYfkT9Bx#scrollTo=hOnaqBjvC_Sc

WORK EXPERIENCE

Smartinsight – Lagos, Nigeria (Remote) *AI Engineer* | July 2023 – Present

- Customize and optimize open-source LLMs (Llama, OpenAI) for business automation tasks, improving response accuracy for internal query systems.
- Developed and integrated AI models into live workflows using Python and Flask, improving decision-making speed for cross-functional teams.
- Cleaned and managed high-volume datasets using SQL and Pandas to ensure quality training data for custom ML models.
- Collaborated on the deployment of scalable AI solutions, focusing on efficient model inference and API performance.

Brattle Multi Concepts Consults – Lagos, Nigeria *Data Scientist & Technical Trainer* | Jan 2022 – July 2023

- Mentored 50+ learners in building ML pipelines, focusing on data analysis and predictive modeling.
- Designed a project-based curriculum covering Python, Scikit-learn, and model evaluation techniques.
- Developed 40+ hours of technical content, including hands-on labs for data visualization and statistical analysis.

EDUCATION

Bachelor of Science (B.Sc.) in Building

Obafemi Awolowo University, Ile-Ife, Osun State, Nigeria

2015 – 2020

Microdegree in Data Science *Ustacky, Lagos, Nigeria* 2021 – 2023

- Coursework: Python for Data Science, Machine Learning, Power BI, SQL, Data Analysis, Capstone Project

CERTIFICATIONS

- **Machine Learning Specialization** – Coursera (Stanford/DeepLearning.AI), 2023
- **Microsoft Power BI Data Analyst Associate** – Microsoft, 2022
- **Machine Learning with Python** – freeCodeCamp, 2021
- **Scientific Computing with Python** – freeCodeCamp, 2021
- **Data Analysis with Python** – freeCodeCamp, 2021
- **Data Analysis with Python** – Coursera (offered by IBM), 2023
- **Databases and SQL for Data Science** – Coursera (offered by IBM), 2023

INTRESTS

AI Agents · Large Language Models (LLMs) · RAG Architecture · AI Ethics · Open Source AI Development